

Revision Notes for Class 12 Physics

Chapter 7 – Alternating Current

Alternating Current

1. The Alternating Current

The magnitude of alternating current changes continuously with time and its direction is reversed periodically. It is represented by

$$I = I_0 \sin \omega t \text{ or } I = I_0 \cos \omega t$$

$$\omega = \frac{2\pi}{T} = 2\pi\nu$$

2. Average Value of Alternating Current

The mean or average value of alternating current over any half cycle is defined as that value of steady current which would send the same amount of charge through a circuit in the time of half cycle (i.e. $T/2$) as is sent by the alternating current through the same circuit, in the same time.

To calculate the mean or average value, let an alternating current be represented by $I = I_0 \sin \omega t$... (1)

If the strength of current is assumed to remain constant for a small time, dt , then small amount of charge sent in a small time dt is

$$dq = Idt \quad \dots (2)$$

Let q be the total charge sent by alternating current in the first half cycle (i.e. $0 \rightarrow T/2$).

$$q = \int_0^{T/2} Idt$$

Using (1), we get, $q = \int_0^{T/2} I_0 \sin \omega t \cdot dt = I_0 \left[-\frac{\cos \omega t}{\omega} \right]_0^{T/2}$

$$= -\frac{I_0}{\omega} \left[\cos \omega \frac{T}{2} - \cos 0^\circ \right]$$

$$= -\frac{I_0}{\omega} [\cos \pi - \cos 0^\circ] \quad (\because \omega T = 2\pi)$$

$$q = -\frac{I_0}{\omega} [-1 - 1] = \frac{2I_0}{\omega} \quad \dots(3)$$

If I_m represents the mean or average value of alternating current over the 1 st half cycle, then

$$q = I_m \times \frac{T}{2} \quad \dots(4)$$

From (3) and (4), we get $I_m \times \frac{T}{2} = 2 \frac{I_0}{\omega} = \frac{2I_0 \cdot T}{2\pi} \quad \dots(5)$

$$\text{or } I_m = \frac{2}{\pi} I_0 = 0.637 I_0$$

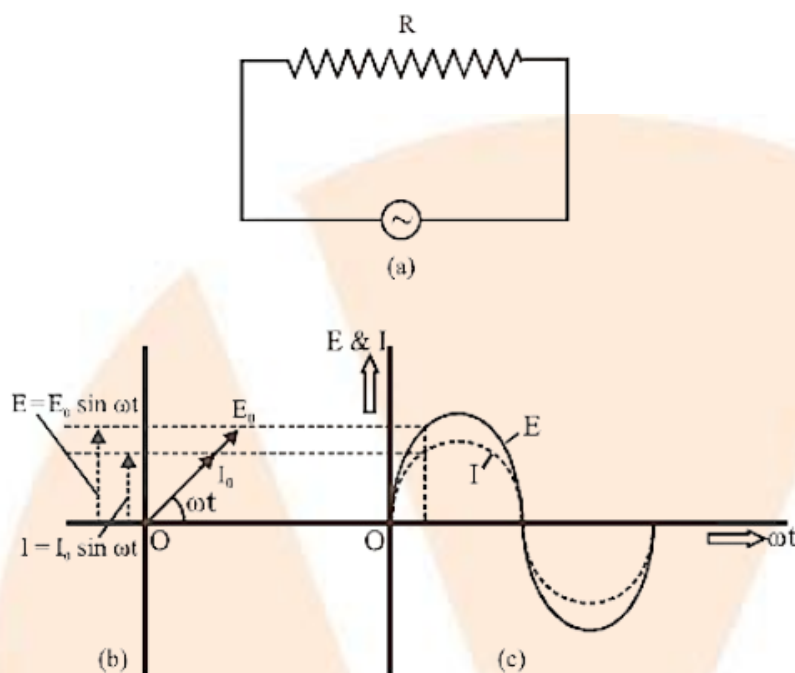
Hence, the mean or average value of alternating current over the positive half cycle is 0.637 times the peak value of alternating current, i.e., 63.7% of the peak value.

3. A.C. Circuit Containing Resistance Only

Let a source of alternating e.m.f. be connected to a pure resistance R , Figure. Suppose the alternating e.m.f. supplied is represented by

$$E = E_0 \sin \omega t \quad \dots(1)$$

Let I be the current in the circuit at any instant t . The potential difference developed across R will be IR . This must be equal to e.m.f. applied at that instant, i.e., $IR = E = E_0 \sin \omega t$



or
$$I = \frac{E_0}{R} \sin \omega t = I_0 \sin \omega t \quad \dots(2)$$

where $I_0 = E_0 / R$, maximum value of current.

This is the form of alternating current developed.

Comparing $I_0 = E_0 / R$ with Ohm's law equation, viz. current = voltage/resistance, we find that resistance to a.c. is represented by R -which is the value of resistance to d.c. Hence behaviour of R in d.c. and a.c. circuit is the same, R can reduce a.c. as well as d.c. equally effectively.

Comparing (2) and (1), we find that E and I are in phase. Therefore, in an a.c. circuit containing R only, the voltage and current are in the same phase, as shown in figure.

3.1 Phasor Diagram

In the a.c. circuit containing R only, current and voltage are in the same phase. Therefore, in figure, both phasors $\vec{I}_0$ and $\vec{E}_0$ are in the same direction making an angle (ωt) with OX. This is so for all times. It means that the phase angle between alternating voltage and alternating current through R is zero.

$$I = I_0 \sin \omega t \text{ and } E = E_0 \sin \omega t$$

4. A.C. Circuit Containing Inductance Only

In an a.c. circuit containing L only alternating current I lags behind alternating voltage E by a phase angle of 90° , i.e., by one fourth of a period. Conversely, voltage across L leads the current by a phase angle of 90° . This is shown in figure.

Figure (b) represents the vector diagram or the phasor diagram of a.c. circuit containing L only. The vector representing $\vec{E}_0$ makes an angle (ωt) with OX. As current lags behind the e.m.f. by 90° , therefore, phasor representing $\vec{I}_0$ is turned clockwise through 90° from the direction of

$$\vec{E}_0 \cdot I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right), I_0 = \frac{V_0}{X_L}, X_L = \omega L$$

A pure inductance offers zero resistance to dc. It means a pure inductor cannot reduce dc. The units of inductive reactance

$$X_L = \omega L \Rightarrow \frac{1}{\text{sec}} (\text{henry}) = \frac{1}{\text{sec}} \frac{1}{\text{amp/sec}} = \text{ohm}$$

The dimensions of inductive reactance are the same as those of resistance.

5. A.C. Circuit Containing Capacitance Only

Let a source of alternating e.m.f. be connected to a capacitor only of capacitance C , figure. Suppose the alternating e.m.f. supplied is

$$E = E_0 \sin \omega t \quad \dots(1)$$

The current flowing in the circuit transfers charge to the plates of the capacitor. This produces a potential difference between the plates. The capacitor is alternately charged and discharged as the current reverses each half cycle. At any instant t , suppose q is the charge on the capacitor. Therefore, potential difference across the plates of capacitor $V = q / C$.

At every instant, the potential difference V must be equal to the e.m.f. applied i.e.,

$$V = \frac{q}{C} = E = E_0 \sin \omega t$$

$$\text{Or } q = C E_0 \sin \omega t$$

If I is instantaneous value of current in the circuit at instant t , then

$$I = \frac{dq}{dt} = \frac{d}{dt}(C E_0 \sin \omega t)$$

$$I = C E_0 (\cos \omega t) \omega$$

$$I = \frac{E_0}{1/\omega C} \sin(\omega t + \pi/2) \quad \dots(2)$$

The current will be maximum i.e.,

$$I = I_0, \text{ when } \sin(\omega t + \pi/2) = \text{maximum} = 1$$

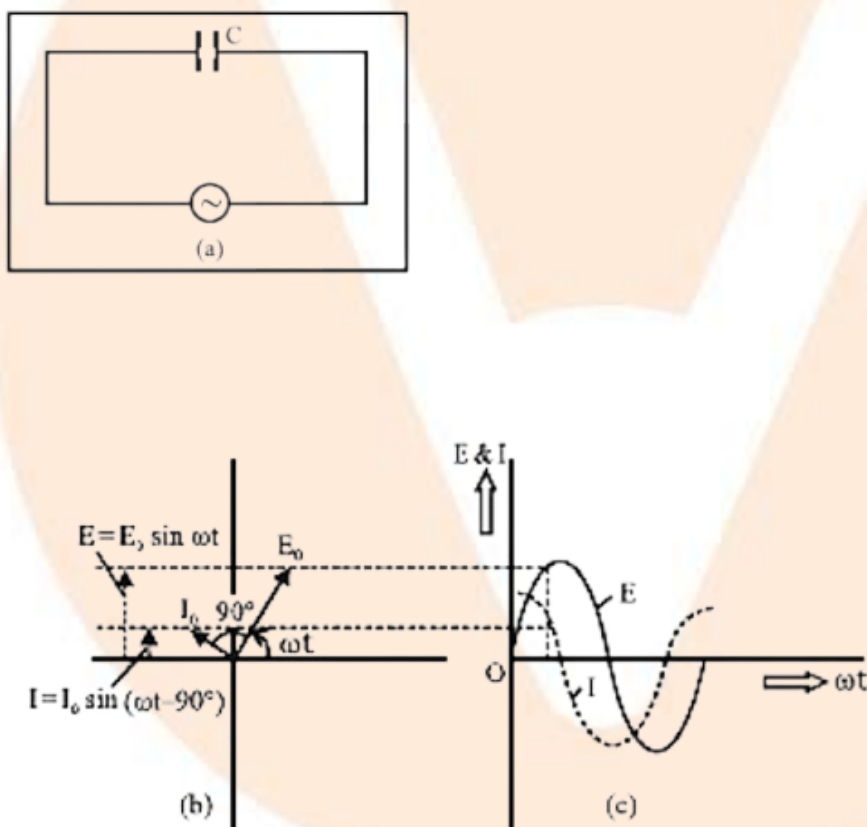
$$\therefore \text{ From (2), } I_0 = \frac{E_0}{1/\omega C} \times 1 \quad \dots(3)$$

$$\text{Put in (2), } I = I_0 \sin(\omega t + \pi/2) \quad \dots(4)$$

This is the form of alternating current developed.

Comparing (4) with (1), we find that in an a.c. circuit containing C only, alternating current leads the alternating e.m.f. by a phase angle of 90° . This is shown in figure (b) and (c).

The phasor diagram or vector diagram of a.c. circuit containing C only is shown in figure (b). The phasor $\vec{I}_0$ is turned anticlockwise through 90° from the direction of phasor $\vec{E}_0$. Their projections on YOY' give the instantaneous values E and I as shown in figure (b). When E_0 and I_0 rotate with frequency ω , curves in figure (c). are generated.



Comparing (3) with Ohm's law equation, viz current = voltage/resistance, we find that $(1/\omega C)$ represents effective resistance offered by the capacitor. This is called capacitive reactance and is denoted by X_c ,

$$\text{Thus, } X_c = \frac{1}{\omega C} = \frac{1}{2\pi VC}$$

The capacitive reactance limits the amplitude of current in a purely capacitive circuit in the same way as the resistance limits the current in a purely resistive circuit. Clearly, capacitive reactance varies inversely as the frequency of a.c. and also inversely as the capacitance of the condenser.

In a d.c. circuit, $\omega = 0, \therefore X_c = \infty$

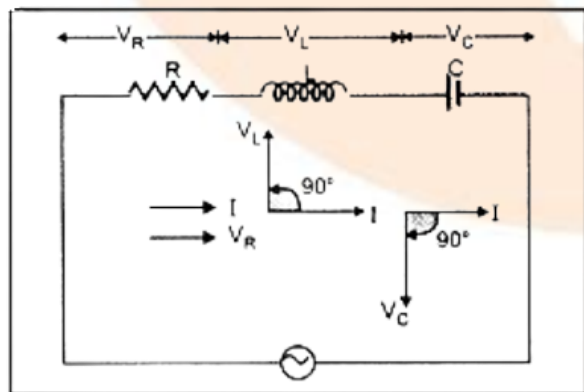
$$X_c = \frac{1}{\omega C} = \text{sec} \frac{1}{\text{farad}} = \frac{\text{sec}}{\text{coulomb/volt}}$$

$$X_c = \frac{\text{voltsec.}}{\text{amp.sec}} = \text{ohm}$$

5. A.C. Circuit Containing Resistance, Inductance and Capacitance and Series

6.1 Phasor Treatment

Let a pure resistance R , a pure inductance L and an ideal capacitor of capacitance C be connected in series to a source of alternating e.m.f. figure. As R, L, C are in series, therefore, current at any instant through the three elements has the same amplitude and phase. Let it be represented by $I = I_0 \sin \omega t$

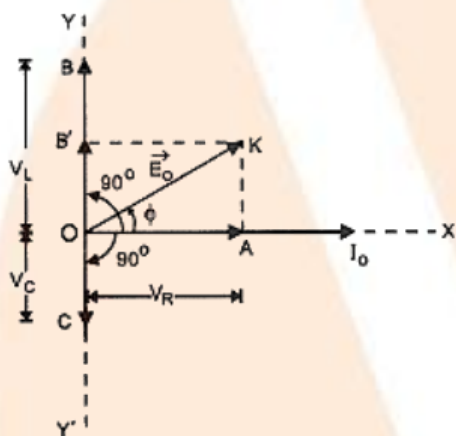


However, voltage across each element bears a different phase relationship with the current. Now,

(i) The maximum voltage across R is

$$\vec{V}_R = \vec{I}_0 R$$

In figure, the current phasor $\vec{I}_0$ is represented along OX.



As $\vec{V}_R$ is in phase with current, it is represented by the vector $\vec{OA}$, along OX.

(ii) The maximum voltage across L is $\vec{V}_L = \vec{I}_0 X_L$

As voltage across the inductor leads the current by 90° , it is represented by $\vec{OB}$ along OY, 90° ahead of $\vec{I}_0$.

(iii) The maximum voltage across C is $\vec{V}_C = \vec{I}_0 X_C$

As voltage across the capacitor lags behind the alternating current by 90° , it is represented by $\vec{OC}$ rotated clockwise through 90° from the direction of $\vec{I}_0$. $\vec{OC}$ is along OY'

6.2 Analytical Treatment of RLC Series Circuit

Let a pure resistance R , a pure inductance L and an ideal condenser of capacity C be connected in series to a source of alternating e.m.f. Suppose the alternating e.m.f. supplied is

$$E = E_0 \sin \omega t \quad \dots(1)$$

At any instant of time t , suppose

q = charge on capacitor

I = current in the circuit

$\frac{dI}{dt}$ = rate of change of current in the circuit

potential difference across the condenser $= \frac{q}{C}$

potential difference across inductor $= L \frac{dI}{dt}$

potential difference across resistance $= RI$

$\therefore$ The voltage equation of the circuit is

$$L \frac{dI}{dt} + RI + \frac{q}{C} = E = E_0 \sin \omega t \quad \dots(2)$$

As $I = \frac{dq}{dt}$, therefore, $\frac{dI}{dt} = \frac{d^2q}{dt^2}$

$\therefore$ The voltage equation becomes

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E_0 \sin \omega t \quad \dots(3)$$

This is like the equation of a forced, damped oscillator. Let the solution of equation (3) be

$$q = q_0 \sin(\omega t + \theta)$$

$$\therefore \frac{dq}{dt} = q_0 \omega \cos(\omega t + \theta)$$

$$\frac{d^2q}{dt^2} = -q_0 \omega^2 \sin(\omega t + \theta)$$

Substituting these values in equation (3), we get

$$L[-q_0 \omega^2 \sin(\omega t + \theta)] + Rq_0 \omega \cos(\omega t + \theta)$$

$$\frac{q_0}{C} \sin(\omega t + \theta) = E_0 \sin \omega t \quad q_0 \omega \left[R \cos(\omega t + \theta) - \omega L \sin(\omega t + \theta) + \frac{1}{\omega C} \sin(\omega t + \theta) \right] = E_0 \sin \omega t$$

As $\omega L = X_L$ and $\frac{1}{\omega C} = X_C$, therefore $q_0 \omega [R \cos(\omega t + \theta) + (X_C - X_L) \sin(\omega t + \theta)] = E_0 \sin \omega t$

Multiplying and dividing by,

$$Z = \sqrt{R^2 + (X_C - X_L)^2}, \text{ we get}$$

$$q_0 \omega Z \left[\frac{R}{Z} \cos(\omega t + \theta) + \frac{X_C - X_L}{Z} \sin(\omega t + \theta) \right] = E_0 \sin \omega t \quad \dots(4)$$

$$\text{Let } \frac{R}{Z} = \cos \phi \text{ and } \frac{X_C - X_L}{Z} = \sin \phi \quad \dots(5)$$

$$\text{So that } \tan \phi = \frac{X_C - X_L}{Z} \quad \dots(6)$$

$$\therefore q_0 \omega Z [\cos(\omega t + \theta) \cos \phi + \sin(\omega t + \theta) \sin \phi] = E_0 \sin \omega t$$

$$\text{or } q_0 \omega Z \cos(\omega t + \theta - \phi) = E_0 \sin \omega t = E_0 \cos(\omega t - \pi/2) \quad \dots(7)$$

Comparing the two sides of this equation, we find that $E_0 = q_0 \omega Z = I_0 Z$, where $I_0 q_0 \omega \dots(8)$

and $\omega t + \theta - \phi = \omega t - \pi / 2$

$$\therefore \theta - \phi = \frac{-\pi}{2}$$

$$\text{or } \theta = \frac{-\pi}{2} + \phi \quad \dots(9)$$

Current in the circuit is

$$I = \frac{dq}{dt} = \frac{d}{dt} [q_0 \sin(\omega t + \theta)] = q_0 \omega \cos(\omega t + \theta)$$

$$I = I_0 \cos(\omega t + \theta) \quad \{ \text{using (8)} \}$$

Using (9), we get, $I = I_0 \cos(\omega t + \phi - \pi / 2)$

$$I = I_0 \sin(\omega t + \phi) \quad \dots(10)$$

$$\text{From (6), } \phi = \tan^{-1} \frac{(X_C - X_L)}{R} \quad \dots(11)$$

As $\cos^2 \phi + \sin^2 \phi = 1$

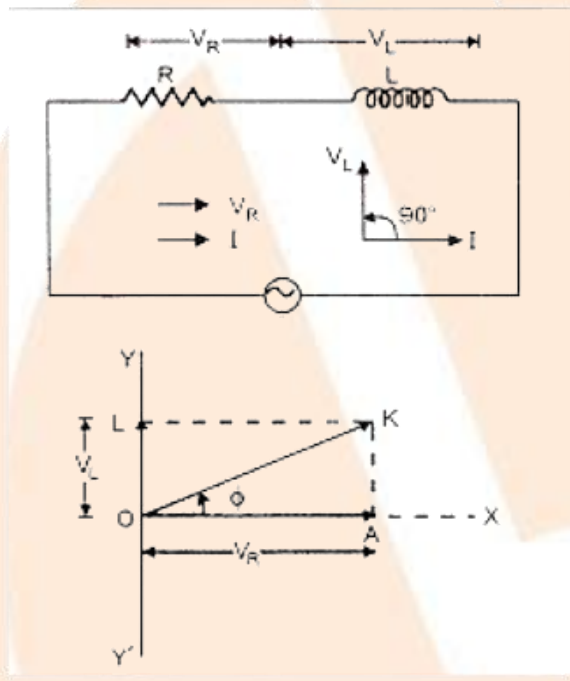
$$\therefore \left(\frac{R}{Z} \right)^2 + \left(\frac{X_C - X_L}{Z} \right)^2 = 1$$

$$R^2 + (X_C - X_L)^2 = Z^2$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} \quad \dots(12)$$

7. A.C. Circuit Containing Resistance & Inductance

Let a source of alternating e m f be connected to an ohmic resistance R and a coil of inductance L , in series as shown in figure.



$$Z = \sqrt{R^2 + X_L^2}$$

We find that in RL circuit, voltage leads the current by a phase angle ϕ , where

$$\tan \phi = \frac{AK}{OA} = \frac{OL}{OA} = \frac{V_L}{V_R} = \frac{I_0 X_L}{I_0 R}$$

$$\tan \phi = \frac{X_L}{R}$$

8. A.C. Circuit Containing Resistance and Capacitance

Let a source of alternating e.m.f. be connected to an ohmic resistance R and a condenser of capacity C , in series as shown in figure.

$$Z = \sqrt{R^2 + X_C^2}$$

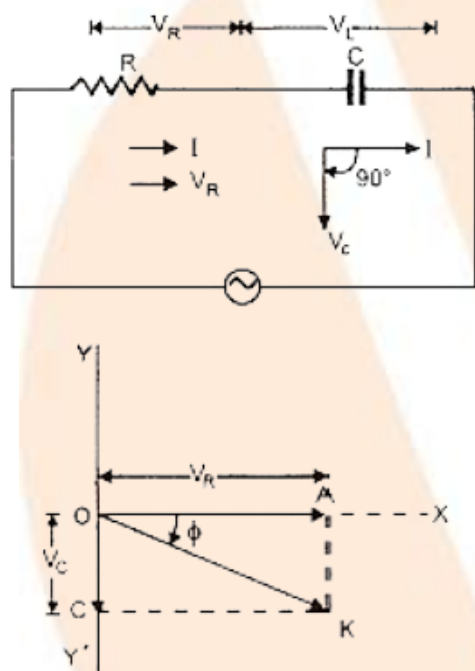


Figure represents phasor diagram of RC circuit. We find that in RC circuit, voltage lags behind the current by a phase angle ϕ , where

$$\tan \phi = \frac{AK}{OA} = \frac{OC}{OA} = \frac{V_C}{V_R} = \frac{I_0 X_C}{I_0 R}$$

$$\tan \phi = \frac{X_C}{R}$$

9. Energy Stored in an Inductor

When a.c. is applied to an inductor of inductance L , the current in it grows from zero to maximum steady value I_0 . If I is the current at any instant t , then the magnitude of induced e.m.f. developed in the inductor at that instant is,

$$E = L \frac{dI}{dt} \quad \dots(1)$$

The self induced e.m.f. is also called the back e.m.f., as it opposes any change in the current in the circuit.

Physically, the self inductance plays the role of inertia. It is the electromagnetic analogue of mass in mechanics. Therefore, work needs to be done against the back e.m.f. E in establishing the current. This work is stored in the inductor as magnetic potential energy.

For the current I at an instant t , the rate of doing work is,

$$\frac{dW}{dt} = EI$$

If we ignore the resistive losses, and consider only inductive effect, then

$$\text{Using (1), } \frac{dW}{dt} = EI = L \frac{dI}{dt} \times I \text{ or } dW = LI dI$$

Total amount of work done in establishing the current I is,

$$W = \int dW = \int_0^I LI dI = \frac{1}{2} LI^2$$

Thus energy required to build up current in an inductor = energy stored in inductor,

$$U_B = W = \frac{1}{2} LI^2$$

10. Electric Resonance

10.1 Series Resonance Circuit

A circuit in which inductance L , capacitance C and resistance R are connected in series, and the circuit admits maximum current corresponding to a given frequency of a.c., is called series resonance circuit.

The impedance (Z) of an RLC circuit is given by

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \quad \dots(1)$$

At very low frequencies, inductive reactance $X_L = \omega L$ is negligible, but capacitive reactance ($X_C = 1/\omega C$) is very high.

As frequency of alternating e.m.f. applied to the circuit is increased, X_L goes on increasing and X_C goes on decreasing. For a particular value of $\omega (= \omega_r, \text{ say })$

$$X_L = X_C$$

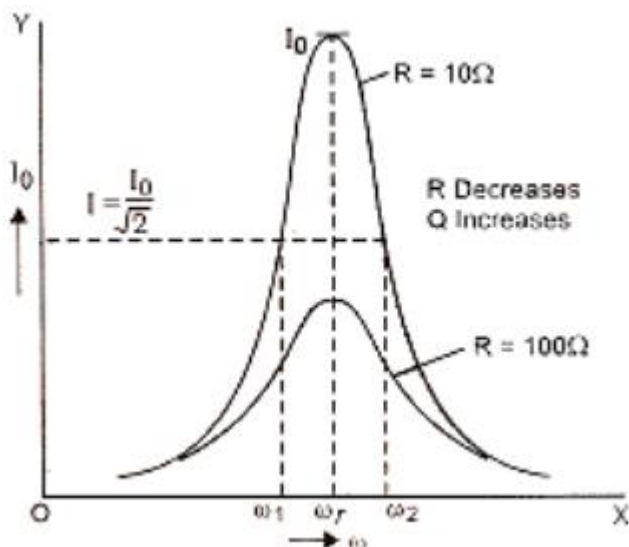
$$\text{i.e., } \omega_r L = \frac{1}{\omega_r C} \text{ or } \omega_r = \frac{1}{\sqrt{LC}}$$

$$2\pi\nu_r = \frac{1}{\sqrt{LC}} \text{ or } \nu_r = \frac{1}{2\pi\sqrt{LC}}$$

At this particular frequency ν_r as $X_L = X_C$, therefore, from (1)

$$Z = \sqrt{R^2 + 0} = R = \text{minimum}$$

i.e. impedance of RLC circuit is minimum and hence the current $I_0 = \frac{E_0}{Z} = \frac{E_0}{R}$ becomes maximum. This frequency is called series resonance frequency.



The Q factor of a series resonant circuit is defined as the ratio of the voltage developed across the inductance or capacitance at resonance to the impressed voltage, which is the voltage applied across R .

i.e.,

$$\text{i.e. } Q = \frac{\text{voltage across } L \text{ or } C}{\text{applied voltage (= voltage across } R \text{)}}$$

$$Q = \frac{(\omega_r L)I}{RI} = \frac{\omega_r L}{R}$$

$$\text{or } Q = \frac{(1/\omega_r C)I}{RI} = \frac{I}{RC\omega_r}$$

Using $\omega_r = \frac{1}{\sqrt{LC}}$, we get

$$Q = \frac{L}{R} \frac{1}{\sqrt{LC}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$\text{or } Q = \frac{1\sqrt{LC}}{RC} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$\text{Thus } Q = \frac{1}{R}\sqrt{\frac{L}{C}} \quad \dots(1)$$

The quantity $\left(\frac{\omega_r}{2\Delta\omega}\right)$ is regarded as a measure of the sharpness of resonance, i.e., Q factor of the resonance circuit is the ratio of resonance angular frequency to bandwidth of the circuit (which is a difference in angular frequencies at which power is half the maximum power or current is $I_0 / \sqrt{2}$).

10.2 Average Power in RLC circuit or Inductive Circuit

Let the alternating e.m.f. applied to an RLC circuit be,

$$E = E_0 \sin \omega t \quad \dots(1)$$

If alternating current developed lags behind the applied e.m.f. by a phase angle ϕ , then

$$I = I_0 \sin(\omega t - \phi) \quad \dots(2)$$

Power at instant t , $\frac{dW}{dt} = EI$

$$\frac{dW}{dt} = E_0 \sin \omega t \times I_0 \sin(\omega t - \phi)$$

$$= E_0 I_0 \sin \omega t (\sin \omega \cos \phi - \cos \omega \sin \phi)$$

$$= E_0 I_0 \sin^2 \omega \cos \phi - E_0 I_0 \sin \omega \cos \omega \sin \phi$$

$$= E_0 I_0 \sin^2 \omega \cos \phi - \frac{E_0 I_0}{2} \sin 2\omega \sin \phi$$

If this instantaneous power is assumed to remain constant for a small time dt , then a small amount of work done in this time is.

$$dW = \left(E_0 I_0 \sin^2 \omega t \cos \phi - \frac{E_0 I_0}{2} \sin 2\omega \sin \phi \right) dt$$

Total work done over a complete cycle is

$$W = \int_0^T E_0 I_0 \sin^2 \omega t \cos \phi dt - \int_0^T \frac{E_0 I_0}{2} \sin 2\omega \sin \phi dt$$

$$W = E_0 I_0 \cos \phi \int_0^T \sin^2 \omega t dt - \frac{E_0 I_0}{2} \sin \phi \int_0^T \sin 2\omega t dt$$

$$\text{As } \int_0^T \sin^2 \omega t dt = \frac{T}{2} \text{ and } \int_0^T \sin \omega t dt = 0$$

$$\therefore W = E_0 I_0 \cos \phi \times \frac{T}{2}$$

$\therefore$ Average power in the inductive circuit over a complete cycle.

$$P = \frac{W}{T} = \frac{E_0 I_0 \cos \phi}{T} \cdot \frac{T}{2} = \frac{E_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cos \phi$$

$$P = E_v I_v \cos \phi \quad \dots(3)$$

Hence average power over a complete cycle in an inductive circuit is the product of virtual e.m.f., virtual current and cosine of the phase angle between the voltage and current.

Note:

The relation (3) is applicable to all a.c. circuits. $\cos \phi$ and Z will have appropriate values for different circuits.

For example:

(i) In PL circuit, $Z = \sqrt{R^2 + X_L^2}$ and $\cos \phi = \frac{R}{Z}$

(ii) In RC circuit, $Z = \sqrt{R^2 + X_C^2}$ and $\cos \phi = \frac{R}{Z}$

(iii) In LC circuit, $Z = X_L - X_C$ and $\phi = 90^\circ$

(iv) In RLC circuit, $Z = \sqrt{R^2 + (X_L - X_C)^2}$ and $\cos \phi = \frac{R}{Z}$

In all a.c. circuits, $I_v = \frac{E_v}{Z}$

10.3 Power Factor of an A.C. Circuit

We have proved that average power/cycle in an inductive circuit is

$$P = E_v I_v \cos \phi \quad \dots(1)$$

Here, P is called true power, (E_v, I_v) is called apparent power or virtual power and $\cos \phi$ is called power factor of the circuit.

$$\text{Thus, Power factor} = \frac{\text{true power (P)}}{\text{apparent power } (E_v I_v)} = \cos \phi \quad \dots(2)$$

$$= \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} \quad [\text{from impedance triangle}]$$

$$\text{Power factor} = \cos \phi = \frac{\text{Resistance}}{\text{Impedance}}$$

In a non-inductance circuit, $X_L = X_C$

$$\therefore \text{Power factor} = \cos \phi = \frac{R}{\sqrt{R^2}} = \frac{R}{R} = 1, \phi = 0^\circ$$

This is the maximum value of power factor. In a pure inductor or an ideal capacitor, $\phi = 90^\circ$.

$$\text{Power factor} = \cos \phi = \cos 90^\circ = 0.$$

Average power consumed in a pure inductor or ideal a capacitor, $P = E_v I_y \cos 90^\circ = \text{Zero}$. Therefore, current through pure L or pure C, which consumes no power for its maintenance in the circuit is called Idle current or Watt less current.

In actual practice, we do not have an ideal inductor or ideal capacitor. Therefore, there does occur some dissipation of energy. However, inductance and capacitance continue to be most suitable for controlling current in a.c. circuits with minimum loss of power.

12. Transformer

A transformer which increases the a.c. voltage is called a step-up transformer. A transformer which decreases the a.c. voltages are called a step-down transformer.

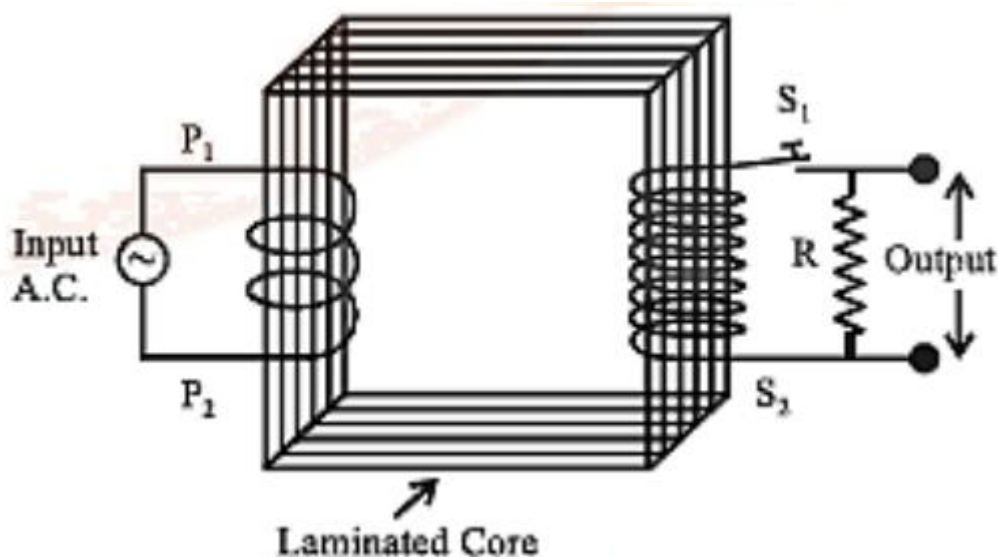
12.1 Principle

A transformer is based on the principle of mutual induction, i.e., whenever the amount of magnetic flux linked with a coil changes, an e.m.f. is induced in the neighbouring coil.

12.2 Construction

A transformer consists of a rectangular soft iron core made of laminated sheets, well insulated from one another, figure. Two coils P_1P_2 (the primary coil) and S_1S_2 (the secondary coil) are wound on the same core, but are well insulated from each other. Note that both the coils are also insulated from the core. The source of alternating e.m.f. (to be transformed) is connected to the primary coil P_1P_2 and a load resistance R is connected to the secondary coil S_1S_2 .

through an open switch S . Thus, there can be no current through the secondary coil so long as the switch is open.



For an ideal transformer, we assume that the resistances of the primary and secondary windings are negligible.

Further, the energy losses due to magnetic hysteresis in the iron core is also negligible. Well-designed high capacity transformers may have energy losses as low as 1%.

12.3 Theory and Working

Let the alternating e.m.f. supplied by the a.c. source connected to primary be

$$E_p = E_0 \sin \omega t \quad \dots(1)$$

As we have assumed the primary to be a pure inductance with zero resistance, the sinusoidal primary current I_p lags the primary voltage E_p by 90° . The primary's power factor, $\cos \phi = \cos 90^\circ = 0$. Therefore, no power is dissipated in primary.

The alternating primary current induces an alternating magnetic flux ϕ_B in the iron core. Because the core extends through the secondary winding, the induced flux also extends through the turns of the secondary.

According to Faraday's law of electromagnetic induction, the induced e.m.f. per turn (E_{turn}) is the same for both, the primary and secondary. Also, the voltage E_p across the primary is equal to the e.m.f. induced in the primary, and the voltage E_s across the secondary is equal to the e.m.f. induced in the secondary. Thus,

$$E_{\text{turn}} = \frac{d\phi_B}{dt} = \frac{E_p}{n_p} = \frac{E_s}{n_s}$$

Here, $n_p; n_s$ represent total number of turns in primary and secondary coils respectively-

$$E_s = E_p \frac{n_s}{n_p} \quad \dots(2)$$

If $n_s > n_p; E_s > E_p$, the transformer is a step up transformer. Similarly, when $n_s < n_p; E_s < E_p$. The device is called a step down transformer. $\frac{n_s}{n_p} = K$ represents transformation ratio.

Note that this relation (2) is based on three assumptions

- (i) the primary resistance and current are small,
- (ii) there is no leakage of magnetic flux. The same magnetic flux links both, the primary and secondary coil,
- (iii) the secondary current is small.

Now, the rate at which the generator/source transfer energy to the primary $= I_p E_p$. The rate at which the primary then transfers energy to the secondary (via the alternating magnetic field linking the two coils) is $I_s E_s$.

As we assume that no energy is lost along the way, conservation of energy requires that

$$I_p E_p = I_s E_s$$

$$\therefore I_s = I_p \frac{E_p}{E_s}$$

From (2),

$$\frac{E_p}{E_s} = \frac{n_p}{n_s}$$

$$\therefore I_s = I_p \frac{n_p}{n_s} = \frac{I_p}{K} \quad \dots(3)$$

For a step up transformer, $E_s > E_p; K > 1 \therefore I_s < I_p$ i.e. secondary current is weaker when secondary voltage is higher, i.e., whatever we gain in voltage, we lose in current in the same ratio.

The reverse is true for a step-down transformer.

$$\text{From eqn. (3) } I_p = I_s \left(\frac{n_s}{n_p} \right) = \frac{E_s}{R} \left(\frac{n_s}{n_p} \right)$$

$$\text{Using equation (2), we get } I_p = \frac{1}{R} \cdot E_p \left(\frac{n_s}{n_p} \right) \left(\frac{n_s}{n_p} \right)$$

$$I_p = \frac{1}{R} \left(\frac{n_s}{n_p} \right)^2 E_p \quad \dots(4)$$

This equation, has the form $I_p = \frac{E_p}{R_{eq}}$, where the equivalent resistance R_{eq} is

$$R_{eq} = \left(\frac{n_p}{n_s} \right)^2 R \quad \dots(5)$$

Thus R_{eq} is the value of load resistance as seen by the source/generator, i.e., the source/generator produces current I_p and voltage E_p as if it were connected to a resistance R_{eq}^{-1} .

Efficiency of a transformer is defined as the ratio of output to the input power.

$$\text{i.e., } \eta = \frac{\text{Output power}}{\text{Input power}} = \frac{E_s I_s}{E_p I_p}$$

In an ideal transformer, where there is no power loss, $\eta = 1$ (i.e. 100%). However, practically there are many energy losses. Hence the efficiency of a transformer in practice is less than one (i.e. less than 100%).

12.4 Energy Losses in a Transformer

Following are the major sources of energy loss in a transformer:

1. Copper loss is the energy loss in the form of heat in the copper coils of a transformer. This is due to Joule heating of conducting wires. These are minimised using thick wires.
2. Iron loss is energy loss in the form of heat in the iron core of the transformer. This is due to the formation of eddy currents in the iron core. It is minimised by taking laminated cores.
3. Leakage of magnetic flux occurs in spite of the best insulations. Therefore, the rate of change of magnetic flux linked with each turn of $S_1 S_2$ is less than the rate of change of magnetic flux linked with each turn of $P_1 P_2$. It can be reduced by winding the primary and secondary coils one over the other.
4. Hysteresis loss. This is the loss of energy due to repeated magnetisation and demagnetisation of the iron core when a.c. is fed to it. The loss is kept to a minimum by using a magnetic material that has a low hysteresis loss.
5. Magnetostriction, i.e., humming noise of a transformer.

Therefore, the output power in the best transformer may be roughly 90% of the input power.